

Caracterizaciones de la convergencia casi segura

Observación 1. Veamos varias caracterizaciones útiles de la convergencia casi segura:

$$\begin{aligned}
X_n \xrightarrow[n \rightarrow \infty]{\text{c.s.}} X &\stackrel{\text{defn}}{\iff} \exists A \in \Sigma : \mathbb{P}(A) = 0 \wedge \forall \omega \in A^c : \lim_{n \rightarrow \infty} X_n(\omega) = X(\omega) \\
&\stackrel{\text{defn de límite en } \mathbb{R}}{\iff} \exists A \in \Sigma : \mathbb{P}(A) = 0 \\
&\quad \wedge \forall \omega \in A^c : \forall \varepsilon > 0 : \exists N \in \mathbb{N} : \forall n \geq N : |X_n(\omega) - X(\omega)| < \varepsilon \\
&\stackrel{\text{liminf-con/1}}{\iff} \exists A \in \Sigma : \mathbb{P}(A) = 0 \wedge \forall \varepsilon > 0 : A^c \subset \liminf_{n \rightarrow \infty} \{|X_n - X| < \varepsilon\} \\
&\iff \forall \varepsilon > 0 : \mathbb{P}\left(\liminf_{n \rightarrow \infty} \{|X_n - X| < \varepsilon\}\right) = 1 \\
&\stackrel{\text{ejer-limsup-liminf-con/1:ejer-limsup-liminf-con/(b)}}{\iff} \forall \varepsilon > 0 : \mathbb{P}\left(\limsup_{n \rightarrow \infty} \{|X_n - X| > \varepsilon\}\right) = 0 \\
&\stackrel{\text{Fatou}}{\iff} \forall \varepsilon > 0 : \limsup_{n \rightarrow \infty} \mathbb{P}(|X_n - X| > \varepsilon) = 0.
\end{aligned}$$